Algorithm Engineering for Policy-Aware CP-SAT Examination Paper Assembly

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Abstract. Sequential examination-paper assembly can be formulated as Boolean constraint satisfaction under structural, temporal, chapter-coverage, and non-duplication requirements. We study algorithmic policy design for a production-derived, metadata-only setting, comparing first-feasible, constructive–local, weighted, direct-compliance, sequential lexicographic, joint lexicographic, and bounded-lookahead policies on a 475-item CBSE Class X Mathematics corpus. Across five independently generated 80-blueprint manifests, sequential $B1$ has seed-level median difficulty deviation 38–40 versus 50–52 for first-feasible $B0$; direct-compliance and weighted policies attain similar difficulty with lower Bloom deviation and runtime. On the fixed manifest, joint and sequential lexicographic policies tie on all 70 comparable cases, while four section orders have identical aggregate medians. Lookahead also ties $B1$ on quality but increases median runtime from 0.124 to 0.257 seconds. Results concern structural alignment and solver runtime, not psychometric validity, end-to-end latency, or universal policy dominance. All reported results are backed by reproducible fixtures, manifests, code, environment, and result files.

Keywords: Constraint Satisfaction Problem · Multi-Objective Optimization · Automated Test Assembly · CP-SAT · Lexicographic Optimization · Educational Assessment.

1 Introduction

Large-scale educational assessment platforms manage repositories containing tens of thousands of test items. Paper assembly must satisfy structural, temporal, pedagogical, and blueprint constraints: exact marks and duration, section-wise counts, cognitive-level balance (Bloom's taxonomy [6, 7]), chapter coverage, and avoidance of duplicate or near-duplicate questions. India's National Education Policy [12] further emphasizes competency-based assessment, making blueprint compliance a first-class engineering requirement.

1.1 The Feasibility Gap

Most production systems treat generation as a Constraint Satisfaction Problem (CSP) and employ SAT or CP solvers configured to terminate at the *first feasible* *solution* ($B0$). While computationally efficient, first-feasible solving exhibits a major flaw: **a feasible paper is not necessarily a high-quality paper**.

Consider a paper requiring M=10 marks and K=4 questions with a target difficulty mix of 2 Easy and 2 Medium items. In a compatible item pool, a baseline first-feasible solver ($B0$) can halt at a valid assignment whose difficulty histogram differs from the target, provided it satisfies the marks, time, count, eligibility, and conflict constraints. An optimized solver ($B1$) instead balances the items to match the target difficulty histogram while maximizing chapter coverage. The gap between an unaligned feasible paper and a blueprint-aligned paper is the *feasibility gap* addressed in this paper. In the present manifest, chapter coverage is nearly saturated; Pass 2 is retained as a policy component but is not empirically differentiating.

1.2 Research Questions and Scope

This work investigates four research questions for a sequential generation setting:

- **RQ1 (Structural Alignment):** How much does multi-objective optimization improve metadata-histogram alignment over first-feasible generation ($B0$), with chapter coverage reported as a saturation check?
- **RQ2 (Solver Runtime):** What solver-runtime overhead does optimization introduce as item pools scale to N = 10,000, excluding pre-solver graph construction?
- **RQ3 (Blueprint Tightness):** How does optimization gain behave as the feasible region contracts under tighter chapter-eligibility restrictions?
- **RQ4 (Soft Compliance):** To what extent does explicit multi-pass optimization improve soft blueprint compliance metrics over first-feasible generation?

Scope and Boundaries. (1) Pre-solver near-duplicate detection constructs a conflict graph S that supplies pairwise at-most-one constraints; duplicate detection is a pipeline boundary condition, not an in-solver objective. (2) Production paper assembly solves template sections sequentially, carrying selected items forward. (3) The evaluation measures structural blueprint alignment and solver-only runtime (model construction and solving); embedding and graph construction costs are excluded, and psychometric IRT validity is not evaluated.

2 Related Work

Automated Test Assembly (ATA) is traditionally formulated as 0–1 integer programming under content and test-information constraints [3, 1, 2, 4]; later work considers parallel-form, cognitive-diagnosis, and multiobjective assembly [14–16]. Our setting is metadata-only, using marks, time, difficulty, chapter, and Bloom labels rather than IRT parameters. Weighted scalarization and explicit lexicographic priorities are standard multiobjective alternatives [8, 9]; we use this distinction to compare policy designs.

Modern CP solvers based on SAT/CDCL techniques, including OR-Tools CP-SAT [10, 5], support Boolean selection, linear constraints, and pairwise conflicts. Our algorithmic focus is the policy layer: sequential lexicographic decomposition, joint optimization as a control, and bounded lookahead around section transitions.

3 Problem Formalization

Let $Q = \{q_1, \ldots, q_n\}$ be an item bank. Each item $q_i = (m_i, t_i, T_i, d_i, b_i)$ has marks $m_i \in \mathbb{Z}^+$, completion time $t_i \in \mathbb{Z}^+$, topic tags $T_i \subseteq \mathcal{T}$, difficulty band $d_i \in \{1, \ldots, |D|\}$, and Bloom's cognitive level $b_i \in \mathcal{B}$. An eligibility set $E \subseteq \mathcal{T}$ filters candidate items: only items satisfying $T_i \cap E \neq \emptyset$ are included in the solver pool. A required-coverage set $R \subseteq \mathcal{T}$ specifies chapters to be represented. In this paper, *chapter coverage* is the instantiated dataset metric $\sum_{t \in R} y_t$, i.e., the number of required chapters represented by at least one selected item; the term is used consistently for this quantity.

A blueprint is specified as $B = (M, W, K, E, R, \Delta, \mathcal{H}, \mathcal{S})$, where M is total marks, W is duration, K is total question count, $\Delta = (\Delta_1, \ldots, \Delta_{|D|})$ is the target difficulty distribution ($\sum_k \Delta_k = K$), and $\mathcal{H}$ and $\mathcal{S}$ represent hard and soft blueprint constraints.

3.1 Decision Variables and Baseline $B0$

Let binary decision variable $x_i \in \{0, 1\}$ represent item selection ($x_i = 1$ iff q_i is selected). The baseline first-feasible solver ($B0$) finds $x \in \{0, 1\}^n$ satisfying:

$$\sum_{i=1}^{n} m_i x_i = M \qquad \text{(Exact total marks)} \qquad (1)$$

$$\sum_{i=1}^{n} t_i x_i = W \qquad \text{(Exact total duration)} \qquad (2)$$

$$\sum_{i=1}^{n} x_i = K \qquad \text{(Exact question count)} \qquad (3)$$

$$x_i + x_j \leq 1 \quad \forall (i, j) \in S \qquad \text{(Anti-duplicate conflicts)} \qquad (4)$$

$$x_i = 1 \Rightarrow T_i \cap E \neq \emptyset \quad \forall i \qquad \text{(Topic eligibility)} \qquad (5)$$

where $S \subseteq Q \times Q$ is the pre-computed set of conflicting near-duplicate pairs. Required chapter coverage is modeled using binary indicators $y_t \in \{0, 1\}$ for $t \in R$:

$$y_t \leq \sum_{i:\, t \in T_i} x_i \quad \forall t \in R \qquad (6)$$

For policies that optimize coverage, this upper-bound linking is sufficient because maximizing y_t forces $y_t = 1$ whenever chapter t is represented. In the reported manifest, coverage is an objective indicator rather than a separate hard requirement; it is not imposed as a mandatory lower bound. For policies that do not optimize coverage, reported coverage is computed directly from the selected item topics rather than from unconstrained y_t values. The baseline $B0$ has no objective function; it halts at the first assignment satisfying the hard constraints.

3.2 Algorithmic Properties

For one section with n_s eligible tier-matched items, e_s conflict edges, and h_s additional hard blueprint rules, the base selection model has n_s item-selection variables and three aggregate equalities, plus e_s conflict inequalities and the additional rules. Eligibility is a domain pre-filter, not an additional solver constraint. In the seven-band, three-Bloom-group implementation, $B1$ adds seven difficulty-deviation variables, $|R|$ coverage indicators, and three Bloom-deviation variables; the absolute-value linearizations contribute two inequalities per difficulty band and two per Bloom group, while coverage contributes $|R|$ linking inequalities. These counts exclude whole-paper cross-section constraints and should be read as model-size components rather than a complete count for every policy. The explicit conflict-constraint count is therefore linear in e_s after graph construction, although solve time need not be linear and constructing S is outside this solver model. The three-pass policy invokes CP-SAT up to three times per section under a 30 s cap; it is a policy layer over the same selection model.

The feasibility problem is NP-hard because cardinality-constrained subset sum is a special case: assign item marks to the input integers, fix count to the required cardinality, set all times to one with $W = K$, and omit conflicts and optional rules. The exact-marks constraint then asks whether a size-K subset sums to M [13]. The count–time dynamic programme used by B_{LS} is therefore pseudo-polynomial in the integer time bound, not a polynomial-time solution to the general feasibility problem.

3.3 Sequential Decomposition: Equivalence Conditions

Let X_s be the items selected before section s. Sequential lexicographic optimization is equivalent to the whole-paper lexicographic model when the problem is separable in the following sense: every lexicographically optimal choice for section s leaves a jointly optimal completion for all later sections, and no objective or constraint couples sections except through the carried exclusions in X_s. This condition can fail when an early optimum consumes items needed later or when cross-section feasibility and soft objectives interact. Hence $B1$ is not claimed lossless in general; B_J tests this decomposition empirically. The observed tie on all 70 common-feasible fixed-manifest cases supports equivalence for the tested instance regime, not in general.

4 Policy Formulations and Optimization Models

We evaluate policy implementations rather than a new solver algorithm. The core comparison contrasts first-feasible search with an independent constructive–local baseline, two single-objective controls, and sequential and joint lexicographic CP-SAT policies under the same blueprint constraints.

4.1 Bounded Lexicographic CP-SAT Policy ($B1$)

$B1$ implements a three-pass lexicographic CP-SAT policy. When every pass returns `OPTIMAL`, it realizes the stated priority order; a time-capped `FEASIBLE` pass instead propagates an incumbent bound and provides only a bounded-time approximation:

1. **Pass 1 (Primary Difficulty Alignment):** Minimize L_1 difficulty deviation $D(x) = \sum_{k=1}^{|D|} u_k$, where auxiliary integer variables $u_k \geq 0$ linearize the absolute error:
$$u_k \geq \sum_{i:\, d_i = k} x_i - \Delta_k \quad \text{and} \quad u_k \geq \Delta_k - \sum_{i:\, d_i = k} x_i \quad \forall k \tag{7}$$
When Pass 1 is `OPTIMAL`, let D^* be its optimal deviation; for a time-capped `FEASIBLE` result, D^* denotes the incumbent deviation.
2. **Pass 2 (Secondary Chapter Coverage):** Constrain $\sum_k u_k \leq D^*$ and maximize required chapter coverage $\sum_{t \in R} y_t$. When Pass 2 is `OPTIMAL`, let C^* be its optimal coverage score; for a time-capped `FEASIBLE` result, C^* denotes the incumbent coverage score.
3. **Pass 3 (Tertiary Soft Compliance):** Constrain $\sum_k u_k \leq D^*$ and $\sum_{t \in R} y_t = C^*$, and minimize Bloom-level deviation $P(x) = \sum_{g \in \mathcal{B}} |\sum_{i: b_i \in g} x_i - T_g|$, where T_g is the target count for cognitive group g.

The implementation performs the three passes sequentially. Pass 1 minimizes total difficulty deviation, Pass 2 maximizes required-chapter coverage subject to the optimal difficulty value, and Pass 3 minimizes Bloom deviation subject to both previous optima. Each pass uses a fresh CP-SAT model with the previous objective value fixed as a constraint; selected item IDs are then carried to the next section.

Status handling. The 30 s budget is a safeguard; all feasible RQ1/RQ4 runs completed below 2.0 s. A `FEASIBLE` timeout satisfies hard constraints but does not certify optimality; such an incumbent is carried to the next pass as a bounded-time approximation. All reported RQ1 common-feasible B1 passes were `OPTIMAL`: 1,050/1,050 pass-level statuses across 70 papers were optimal (artifact: `results_b1_status_audit.json`).

4.2 Alternative Baseline and Control Policies

- B_{LS} **(Constructive–Local Baseline):** A non-CP-SAT constructive dynamic programme selects an exact count–time seed using greedy metadata priorities, followed by deterministic improving equal-time swaps. The dynamic programme enforces hard feasibility; its scoring and local search do not optimize the CP-SAT objectives.

- $B2$ **(Weighted-Sum Scalarization):** Solves a single CP-SAT pass minimizing $\alpha D(x) + \beta U(x) + \gamma P(x)$, where $U(x) = |R| - \sum_{t \in R} y_t$. The default shared-suite weight vector is $(\alpha, \beta, \gamma) = (1, 1, 1)$, using raw rather than normalized objective components; results should therefore be interpreted as specific to this policy scale.
- B_C **(Direct Compliance Control):** A joint whole-paper CP-SAT model minimizes $D_{\text{total}} + P_{\text{paper}}$, the affine equivalent of the reported complete-paper soft-compliance loss because every template paper has 38 questions. It is a single-objective control, not a new optimization method.
- B_J **(Joint Lexicographic Control):** A whole-paper CP-SAT model applies the same $D \to C \to P$ priority order as $B1$, while enforcing all section constraints and cross-section uniqueness simultaneously.

4.3 Bounded Lookahead-Guided Sequential Diagnostic

To test whether sequential lock-in can be reduced without changing the production decomposition, we add a bounded-lookahead policy. At each section it generates up to $k = 3$ distinct $B1$ candidates with
$varepsilon = 0$ tolerance and scores their downstream states using a coarse relaxed estimator. Let $P_s(x)$ be the eligibility- and mark-tier-filtered pool for a remaining section s after candidate x is excluded, and let r_s, w_s be its required count and time. Define the binary reachability term

$$R_s(x) = \mathbf{1}\left\{ \sum_{q \in P_s(x)} 1 \geq r_s \right\}, \tag{8}$$

and the time-slack term

$$\sigma_s(x) = \left| \sum_{q \in P_s(x)} t_q - w_s \right|. \tag{9}$$

The candidate score is

$$S(x) = 100 \sum_s R_s(x) - \sum_s \sigma_s(x), \tag{10}$$

with the highest-scoring candidate selected. The estimator is not a completion counter and does not certify future feasibility; the $+100/-100$ scale is a transparent diagnostic rather than a fitted parameter. Each candidate-generation subsolve uses one worker, seed 1, and a 5 s cap. On the fixed manifest, lookahead adds roughly a factor-of-two runtime overhead without improving quality, so it is treated as a secondary ablation rather than a production recommendation.

4.4 Constructive–Local Baseline Details

The constructive–local baseline uses a count–time dynamic program to build a feasible section, then applies deterministic equal-time swaps that improve

$(D, P, -C)$ while preserving conflicts. Its state complexity is $O(n_s KW)$ without conflicts, with an additional implementation factor for conflict checks. Unlike CP-SAT policies, it is heuristic and provides no optimality certificate.

4.5 Compliance Metrics

Blueprint compliance is partitioned into non-negotiable hard rules $\mathcal{H}$ (e.g., exact marks, time, question count) and soft distributional goals $\mathcal{S}$ (difficulty alignment and Bloom fit). For hard compliance, $CR_{\text{hard}}(x) = 1.0$ for any valid output. Soft compliance is scored as:

$$CR_{\text{soft}}(x) = \frac{1}{2}\left(\left[1 - \min\left(1, \frac{D(x)}{2K}\right)\right] + \left[1 - \min\left(1, \frac{P(x)}{2K}\right)\right]\right) \tag{11}$$

For a section, $D = \sum_{k=1}^{7} |\sum_{i:d_i=k} x_i - \Delta_k|$ and $P = \sum_{g\in\{\text{RU,APP,AEC}\}} |\sum_{i:b_i\in g} x_i - T_g|$; because both observed and target histograms sum to K, $0 \le D, P \le 2K$. For a complete template paper, $K_{\text{tot}} = 38$, D_{total} is the sum of section deviations, P_{paper} uses the 54/24/22 paper-level target, and $0 \le D_{\text{total}}, P_{\text{paper}} \le 76$. Hence the clipping in CR_{soft} is inactive here and $CR_{\text{soft}} = 1 - (D_{\text{total}} + P_{\text{paper}})/(4K_{\text{tot}})$. For $B2$, $U = |R| - \sum_{t\in R} y_t \in [0, 7]$ and the fixed raw objective is $D + U + P$; its unnormalized $(1, 1, 1)$ weights were chosen before all runs as a concrete scalar policy, not tuned from the sensitivity grid. It is therefore a different preference model from lexicographic $B1$, not a direct test of $B1$'s priority semantics.

5 Experimental Evaluation

5.1 Dataset and Benchmark Setup

The empirical corpus consists of **475 CBSE Class X Mathematics items** retrospectively extracted from a production assessment system. Every item contains populated metadata: marks (1–5), completion time, ordinal difficulty (bands 1–7), Bloom cognitive tag, and chapter classifications. The study does not evaluate a live deployment, user adoption, or operational outcomes. The bank is easy-skewed (83% of items in bands 1–3) and marks-skewed (73% 1-mark items), providing a production-derived test setting for difficulty alignment. For solver scaling experiments (RQ2), synthetic pools calibrated to the empirical marginal distributions are generated up to N=10,000; conflict density and feature correlations are not claimed to reproduce the production bank.

5.2 Blueprint Manifest Generation

For RQ1, 80 blueprints are deterministically generated using a fixed random seed (`random.Random(42)`). The seven required chapters form the chapter set for each blueprint; chapter labels are represented through the topic-tag metadata, and the eligibility sample is the blueprint-specific subset of those chapters. For

Table 1. RQ1 (Table 3): outcomes across the 70 blueprints feasible for all six core policies in the fixed 80-blueprint manifest. All reported performance entries are medians with IQRs in brackets, except coverage, which is a median count. D: sum of five section-level difficulty deviations (lower is better); coverage: raw number of represented required chapters ($|R| = 7$); Bloom: complete-paper L_1 deviation from the 54/24/22 split. Solver runtime is model construction plus solving for a complete paper, excluding embedding and conflict-graph construction; lower runtime is better. The 10 exclusions are jointly infeasible under modeled hard constraints.

Model	**Feasible**	$D\downarrow$	**Coverage**	**Bloom**$\downarrow$	**Solver s**
$B0$ first-feasible	70/80	52 [48,56]	6	40 [40,40]	0.03
B_{LS} constructive–local	70/80	42 [38,44]	6	**22** [22,26]	0.73
$B2$ weighted-sum	70/80	**40** [36.5,44]	6	**22** [22,26]	0.07
B_C direct compliance	70/80	**40** [38,44]	6	**22** [22,24]	**0.04**
B_J joint lexicographic	70/80	**40** [36,44]	6	24 [22,26]	0.17
$B1$ sequential lexicographic	70/80	**40** [36,44]	6	24 [22,26]	0.13

each ID in ascending order, 5–7 eligible chapters are sampled uniformly from the sorted chapter list; target difficulty counts are normalized draws weighted $(4, 5, 4, 3, 2, 1, 0.3)$ with cyclic remainder assignment, and Bloom targets use the 54/24/22 split. Each blueprint specifies five template sections (A–E), containing 38 questions in total, with section-level question counts, marks, time bounds, and target difficulty distributions. The fixed structural parameters used for every manifest blueprint are: A/B/C/D/E contain 20/5/6/4/3 questions with 20/10/18/20/12 marks and 40/15/36/40/18 minutes; the seed-42 section-level target histograms and Bloom-group targets are recorded in the blueprint manifest and regeneration artifacts. The paper-level Bloom target is 54/24/22; section-level targets are generated with the manifest rather than assumed constant across sections. Sections are solved sequentially in order A→E, carrying selected item IDs forward to enforce cross-section uniqueness.

5.3 Whole-Paper Control on Excluded Blueprints

A whole-paper joint CP-SAT audit classifies all 10 sequentially excluded blueprints as `INFEASIBLE` under the same hard constraints, indicating global blueprint infeasibility rather than sequential lock-in; the common-feasible baseline is therefore 70/80.

5.4 Results

RQ1 (Structural Alignment). The fixed manifest provides paired policy comparisons. On the fixed manifest, $B1$ improves D over $B0$ in all 70 pairs (median difference 12, 95% CI [10, 12], Holm-adjusted $p = 3.6{\times}10^{-12}$, $r = 0.87$). These are paired results for the tested manifest, not independent evidence from 70 unrelated experiments. Against B_{LS}, $B1$ improves D in 22 pairs and ties in 48 (median difference 0, CI [0, 0], adjusted $p = 1.4{\times}10^{-5}$, $r = 0.56$). This establishes structural-alignment differences only, not psychometric or pedagogical superiority.

Table 2. Secondary bounded lookahead diagnostic (Table 4) on the fixed 80-blueprint manifest. All entries are medians with IQRs in brackets; for quality metrics, positive values in the final column favour lookahead, while the runtime difference is defined as Sequential $B1$ minus Lookahead, so the negative runtime value directly indicates that lookahead is slower.

Metric	Sequential $B1$	Lookahead k=3	Difference ($B1$−lookahead)
Difficulty D	40 [36,44]	40 [36,44]	0
Bloom deviation	24 [22,26]	24 [22,26]	0
Chapter coverage	6 [6,7]	6 [6,7]	0
Solver runtime (s)	0.124 [0.109,0.140]	0.257 [0.226,0.288]	-0.132

The direct-compliance control B_C and weighted $B2$ match $B1$'s median difficulty while achieving lower median Bloom deviation (22 versus 24) and lower solver runtime (0.04 and 0.07 versus 0.13 s). Because the theoretical bounds make clipping inactive for every feasible paper, and every complete template paper contains 38 questions, $CR_{\text{soft}} = 1 - (D_{\text{total}} + P_{\text{paper}})/(4K_{\text{tot}})$ differs from minimizing $D_{\text{total}} + P_{\text{paper}}$ only by a positive affine transformation; this makes B_C affine-equivalent to the reported complete-paper loss, but not equivalent to $B1$, whose lexicographic $D \to C \to P$ preference protects each priority level sequentially rather than trading the components in a single sum. Thus $B1$ is not presented as the raw quality–runtime optimum: its contribution is a deterministic, weight-free specification of priority order. Coverage has limited discriminating power in this template: all paired core-policy comparisons tie in represented-chapter count, despite a per-policy IQR of [6,7] over seven possible chapters. It is therefore not evidence that Pass 2 improves quality.

Baseline sensitivity. These are fixed-policy comparisons, not universal comparisons. $B0$ uses one worker, the documented deterministic search configuration, and the per-blueprint seeds retained in the artifact; this study does not estimate a full solver-seed distribution. $B2$ uses the pre-specified raw $(1, 1, 1)$ weights in Table 1. As a secondary sensitivity check, all 27 raw vectors in $\{1, 3, 10\}^3$ retain median chapter coverage 6 and yield median $D \in \{40, 41\}$, median Bloom deviation $\in \{22, 23, 24\}$, and median solver runtime 0.072–0.126 s on the same 70 common-feasible rows (artifact: `results_weight_sensitivity.json`). This bounded grid demonstrates policy-scale sensitivity but does not select an optimal weight vector or convert $B2$ into a priority-policy comparison.

The bounded lookahead diagnostic ties sequential $B1$ on difficulty, Bloom deviation, and coverage across all 70 comparable fixed-manifest papers, while increasing median solver runtime from 0.124 to 0.257 s (an increase of 0.132 s; lower runtime is better). It therefore provides a reproducible test of downstream-aware candidate selection, but no evidence of a quality gain on this corpus; the observed overhead makes it unsuitable as the default production policy here.

Robustness, Joint Control, and Ablations. The primary robustness evidence is the seed-level paired analysis across five independently generated manifests (seeds 42–46), with the manifest treated as the resampling unit. Across these five clusters, $B1$ has seed-level median D values of 38–40 versus 50–52 for $B0$. The corresponding paired effects, cluster-bootstrap 95% intervals, effect sizes, ties, and Holm-adjusted paired tests are reported at the seed level in `results_multiseed_revision.json`. The 341 pooled common-feasible blueprint rows are retained only as a descriptive aggregation; they are not treated as 341 independent replicates because blueprints within a manifest share generation rules and a random seed. The same reporting convention is used for the $B1$ versus B_{LS} comparison: the pooled 77/264/0 improve/tie/lose counts are descriptive, while inference is based on the clustered seed-level paired effects.

On the fixed manifest, B_J and sequential $B1$ tie on difficulty, Bloom deviation, and coverage in all 70 jointly feasible cases; B_J is slower (median 0.17 versus 0.13 s). Production order A→E, reverse order, and high-/low-count-first orders each retain 70 feasible cases and identical aggregate medians. Hence no ordering loss was observed for the evaluated corpus and template; this is not evidence that sequential assembly is lossless in general. Full-suite objective ablations give median (D, Bloom) of $(40, 32)$ for D-only, $(40, 38)$ for D→C, $(40, 24)$ for D→C→P, and $(48, 22)$ for C→P; chapter coverage again has median 6. These are descriptive component effects over the 70 common cases, not independent replicates.

RQ2 (Solver Scaling). For a Section-A blueprint (K=20), median solver runtime rises from 0.003/0.021 s for $B0/B1$ at $N = 100$ to 0.326/1.643 s at $N = 10{,}000$. $B0$ executes in under 0.33 s at $N = 10{,}000$, while $B1$ takes 1.64 s (a 2–7× overhead across the displayed sizes). Under the stated synthetic calibration, all runs remained within budget; this does not establish end-to-end production latency. A descriptive log–log fit over the five displayed pool sizes gives slopes of approximately 1.00 for $B0$ and 0.93 for $B1$. These are empirical slopes over this calibrated range, not asymptotic complexity claims; conflict-graph construction, embedding, memory, and larger section sizes remain outside this experiment.

RQ3 (Blueprint Tightness). In a separate tightness sweep of 100 blueprint–tier cases in total (20 cases per tier), restricting eligible chapter counts from 7 down to 3, common B0/B1 feasibility drops from 20/20 (tier 7) to 5/20 (tier 3) in the sweep. Conditional on common feasibility, $B1$ maintains a median difficulty-deviation improvement of 6 to 12 units over $B0$ across the tested tiers; these cases are treated as a sensitivity analysis rather than as independent replications. The runner records common-feasible versus non-common-feasible cases, but not policy-specific `INFEASIBLE/UNKNOWN` categories for this legacy sweep; consequently, the table is an outcome accounting for the paired estimand, not evidence of a policy-specific feasibility advantage.

RQ4 (Soft Compliance). Each pair consists of the B0 and B1 outputs for the same generated blueprint and eligibility tier; pairs are retained only when both policies return feasible outputs. Among 66 common-feasible blueprint pairs retained from the tightness sweep, using the complete-paper form of CR_{soft} rather than an average of section scores, $B1$ increases median soft compliance score from 0.408 ($B0$) to 0.566 and outperforms $B0$ in all 66 cases (Wilcoxon $p = 1.6{\times}10^{-12}$, $r = 0.87$). Because the analysis uses common-feasible pairs, these results should be interpreted together with the tier-wise feasibility counts rather than as a feasibility advantage.

5.5 Reproducibility

The public reproducibility repository is `https://github.com/papudg/Research`, branch `reproducibility-public`, pinned for this submission at commit `cd25c2d90c5d48df2817c7bd2fb446355d4179b4`. It contains the harness in `experiments/`, library source in `src/icaa/`, dependency pins in `requirements.txt`, fixed manifests under `data/manifests/`, and recorded results. The metadata-only 475-record fixture is at `docs/qbank_audit/cbse_class_x_maths_questions_full_updated.json` (no question text in the public release). From the repository root, set `PYTHONHASHSEED=0` and run `python experiments/revision_report.py`; the platform-specific exact command list is in `REPRODUCIBILITY.md`. `results_environment.json` records the CPU, RAM, Windows, Python/OR-Tools versions, and fixed one-worker CP-SAT parameters. Runtime results remain solver/model-construction measurements, not end-to-end preprocessing latency.

The public artifact maps each main claim to recorded JSON/text results: RQ1 to `results_revision_suite.json`, lookahead to `results_lookahead_comparison.json`, clustered analysis to `results_multiseed_revision.json`, joint/order controls to `results_joint_policy_comparison.json`, tightness to `results_rq3_rq5.txt`, weight sensitivity to `results_weight_sensitivity.json`, and environment settings to `results_environment.json`.

6 Discussion and Limitations

Policy Design Trade-Offs. First-feasible generation leaves structural alignment uncaptured relative to optimized controls. Multi-pass $B1$ supplies an explicit, weight-free priority hierarchy: difficulty alignment first, chapter coverage second, and Bloom fit third. Its extra passes are therefore a methodological and governance choice rather than a runtime-optimal policy. On this corpus, $B1$ is about twice as slow as the fastest optimized controls and gains no lookahead quality, so a different latency budget may favor B_C or $B2$.

Limitations. Structural alignment does not imply psychometric validity; runtimes exclude embedding and conflict-graph construction. The evaluation uses one 475-item CBSE Class X Mathematics corpus and fixed template family, so cross-subject, cross-bank, and cross-template generalization is untested. Difficulty and Bloom labels are source-system metadata, the upstream duplicate detector was not human-validated, and fixed solver settings/weights make sequential–joint agreement an instance-regime result rather than a universal guarantee.

7 Conclusion

We presented an algorithm-engineering framework for policy-aware CP-SAT examination-paper assembly, with explicit comparisons of sequential and joint optimization. Across five manifests, sequential lexicographic $B1$ has median difficulty deviations of 38–40 versus 50–52 for first-feasible $B0$. On the fixed manifest, $B1$ and joint B_J tie on all 70 comparable cases, while alternative section orders retain the same aggregate medians. Bounded lookahead likewise provides no quality gain and increases median runtime from 0.124 to 0.257 seconds. These results support sequential optimization as an effective policy for the tested instance regime while leaving end-to-end latency, psychometric validity, and cross-corpus generalization for future work.

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